Polygons & Quadrilaterals

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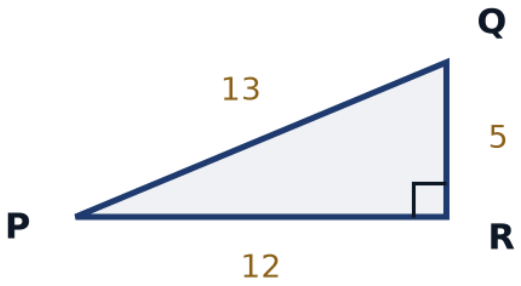
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

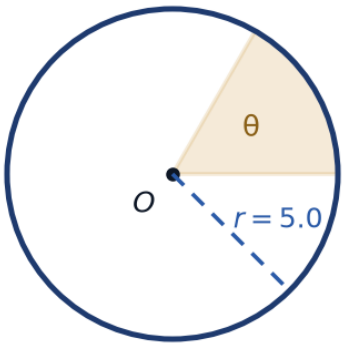
2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the area of a circle with radius 5.



4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A solid metal cone with base radius 4 cm and height 9 cm is submerged into a cylindrical beaker with internal radius 6 cm partially filled with water. Assuming the cone is fully submerged and no water overflows, by how many centimeters does the water level in the beaker rise?

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Let $f(x) = \frac{2x-7}{5}$. (a) Find $f^{-1}(x)$. (b) Solve the equation $f(f^{-1}(x)) = 9$. (c) Find the value of k for which $f(k) = f^{-1}(k)$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find all real solutions to the equation $|2x - 7| = 3x - 1$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A parabola has its vertex at $(3, -4)$ and passes through the point $(1, 0)$. (a) Write the equation of the parabola in standard form $y = ax^2 + bx + c$. (b) Determine the coordinates of the other x-intercept.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times$ IQR rule to determine whether 72 is an outlier.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An equilateral triangle has a side length of 12 cm. Find: (a) the exact length of its altitude, and (b) its exact area.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two assembly line components operate independently. Component A has a 95% reliability rate (probability of functioning properly is 0.95), and Component B has a 90% reliability rate (probability of functioning is 0.90). What is the probability that at least one of the two components fails?

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of the constant a will the system of equations $ax + 4y = 7$ and $3x + 2y = 5$ have no solution?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Determine the maximal real domain of the function $f(x) = \frac{\sqrt{3x-12}}{x^2-25}$. Express your answer in interval notation.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $x - 2$ is a factor of the polynomial $P(x) = x^3 - kx^2 + 3x - 10$, find the value of k, and determine all real roots of $P(x)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Coordinate Geometry

A rectangular water storage reservoir has dimensions 2.5 meters by 3 meters by 1.2 meters. A drainage pump empties the reservoir at a constant rate of 75 liters per minute. Given that $1\ \text{m}^3 = 1000\ \text{liters}$, how many hours will it take to completely empty the full reservoir?

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Probability

In the system $ax + 4y = 18$ and $3x + 2y = 9$, the system has no solution. What is the value of a?

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Lines & Angles

In triangle OAB, $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$. Point P lies on AB such that $AP : PB = 1 : 2$. Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Triangles & Congruence

A line segment connects points $P(-3, 7)$ and $Q(5, -1)$. Determine: (a) the exact length of PQ, (b) the coordinates of the midpoint M, and (c) the slope of the perpendicular bisector of PQ.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

The graph of $y = f(x)$ passes through the points $(-2, 8)$ and $(4, -6)$. If $g(x) = 3f(x - 5) + 2$, determine the corresponding points on the graph of $g(x)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

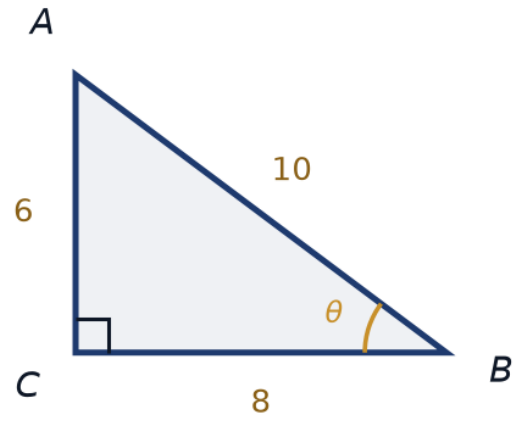
Solve the quadratic equation $3x^2 + 10x - 8 = 0$ by factoring.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 99°

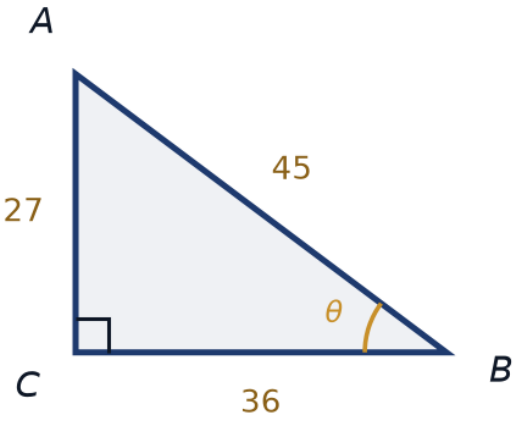
b. 109°

c. 91°

d. 144°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying review: identifying similar triangles, $m\angle A = 37^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 98°

b. 108°

c. 90°

d. 143°

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23 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In parallelogram $ABCD$, $\angle A = (3x + 4)^\circ$ and $\angle B = (5x + 16)^\circ$. Find $\angle A$.



a. 116°

b. 64°

c. 74°

d. 20°

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In parallelogram $ABCD$, $\angle A = 132^\circ$. What is the measure of $\angle B$?



a. 228°

b. 132°

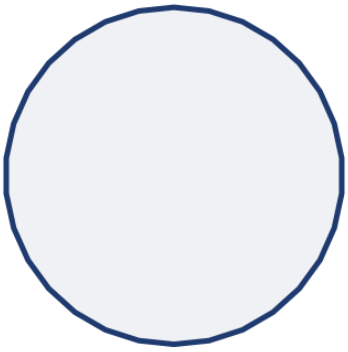
c. 48°

d. 42°

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the measure of one exterior angle of a regular 30-gon?



a. 12°

b. 348°

c. 168°

d. 24°

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The diagonals of a parallelogram bisect each other. If one diagonal has length 20, what is the length of half of that diagonal?



a. 10

b. 20

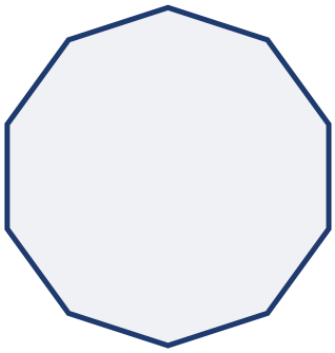
c. 12

d. 40

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

How many diagonals does a convex polygon with 10 sides have?



a. 35

b. 34

c. 10

d. 45

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A trapezoid has parallel bases of length 28 and 28. What is the length of the midsegment?



a. 28

b. 56

c. 31

d. 0

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In isosceles trapezoid $ABCD$, base AB is parallel to base CD with $AB > CD$. The midsegment MN connects the midpoint M of leg AD to the midpoint N of leg BC. Diagonals AC and BD intersect MN at points P and Q, respectively. If $MN = 16$, $PQ = 5$, and the leg length is $AD = 13$, what is the length of diagonal AC?

a. 20

b. 18

c. $5\sqrt{17}$

d. $2\sqrt{61}$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In rhombus $ABCD$, the diagonals AC and BD intersect at point E. If $m\angle BAC = (3x + 10)^\circ$ and $m\angle ABD = (2x - 5)^\circ$, what is the measure of $\angle ABC$?

a. 29°

b. 58°

c. 61°

d. 122°

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Four interior angles of a convex pentagon have measures of 110°, 120°, 95°, and 130°. What is the measure of the fifth interior angle?

a. 75°

b. 85°

c. 95°

d. 105°

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the rhombus ABCD shown below, the diagonals AC and BD intersect at point E. If the length of diagonal AC is $10x$ units and the length of diagonal BD is $12x$ units, and the perimeter of the rhombus is 52 units, what is the length of side AB?

a. 13 units

b. 26 units

c. 10 units

d. 12 units

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A regular hexagon has interior angles that are all equal. What is the measure of each interior angle of a regular hexagon?

a. 100°

b. 108°

c. 120°

d. 150°

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the sum of the interior angles of a polygon with 25 sides?

a. 4320°

b. 4500°

c. 3960°

d. 4140°

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two similar polygons have corresponding sides in the ratio $3 : 8$. What is the ratio of their areas?

a. 3 : 8

b. 6 : 16

c. 9 : 64

d. 27 : 512

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a regular polygon, each interior angle is 1 times its corresponding exterior angle. How many sides does the polygon have?

a. 3

b. 8

c. 6

d. 4

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A rectangle has length 3 and width 4. Find the length of its diagonal.

a. 13

b. 12

c. 7

d. 5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Triangles & Congruence

A rectangle has diagonal 5 and length 3. Find the width.

a. 6

b. 8

c. 2

d. 4

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A trapezoid has parallel sides 6 and 10 with height 4. Find its area.

a. 35

b. 24

c. 32

d. 64

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A rectangle has perimeter 38 and length 10. Find the width.

a. 28

b. 19

c. 9

d. 11

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Circles

A parallelogram has base 9 and height 5. Find its area.

a. 28

b. 14

c. 54

d. 45

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Area, Surface Area & Volume

The sum of interior angles of a convex polygon is 360°. How many sides does it have?

a. 7

b. 5

c. 4

d. 2

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Transformations & Symmetry

A rectangle has area 48 and one side 6. Find the other side.

a. 42

b. 6

c. 10

d. 8

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Foundations & Reasoning

A rectangle has length 9 and width 4. Find its perimeter.

a. 13

b. 28

c. 36

d. 26

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Coordinate Geometry

A rhombus has side length 6. Find its perimeter.

a. 36

b. 28

c. 24

d. 12

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